

Limits

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I Introduction

Before we formally define a limit, we must define a cluster point.

Definition. Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is a **Cluster Point** of A if for every $\delta > 0$ there exists at least one point $x \in A, x \neq c$ such that $|x - c| < \delta$.

In other words, a point c is a cluster point of the set A if every δ neighborhood of $V_\delta(c) = (c - \delta, c + \delta)$ of c contains at least one point of A distinct from c . Consider $A = \left(0, \frac{1}{2}\right)$. The cluster points of A are $\left(0, \frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \dots\right)$. Now we are ready to define the limit.

Definition. Let $A \subseteq \mathbb{R}$, and let c be a cluster point of A . For a function $f : A \rightarrow \mathbb{R}$, a real number L is said to be the **limit of f at c** if, for all $\epsilon > 0$ there exists $\delta > 0$ such that for all $x \in A$ and $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

In other words, the limit of $f(x)$ is L if for every ϵ neighborhood $V_\epsilon(L)$ of L , there exists a δ neighborhood $V_\delta(c)$ around c with the property that for all $x \in V_\delta(c)$ different from c , (and with $x \in A$), it follows that $f(x) \in V_\epsilon(L)$.

Examples

1. Prove that

$$\lim_{x \rightarrow 1} \frac{x+1}{2-x} = 2.$$

Proof.

$$\forall \epsilon > 0, \text{ find } \delta(\epsilon) \text{ s.t. } \forall x \in A, 0 < |x - 1| < \delta.$$

Then,

$$\left| \frac{x+1}{2-x} - 2 \right| < \epsilon \implies \left| \frac{x+1-2(2-x)}{2-x} - 2 \right| < \epsilon \implies \left| \frac{x+1-4+2x}{2-x} - 2 \right| < \epsilon$$

[†]Definitions and example problems from MATH 42001: Analysis I instructed by Dr. Artem Zvavitch, *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert, and *Understanding Analysis* by Stephen Abbott

$$\implies \left| \frac{3|x-1|}{2-x} - 2 \right| < \epsilon \implies \left| \frac{x-1}{2-x} - 2 \right| < \epsilon$$

Assume $\delta \leq \frac{1}{2}$, then $|x-1| \leq \frac{1}{2} \implies \frac{1}{2} < x < \frac{3}{2}$ so $|2-x| < 2$. Now we have, $\delta \cdot 3 \cdot 2 \leq \epsilon$, so pick $\delta = \frac{\epsilon}{6}$. Therefore, $\delta = \min \left\{ \frac{1}{2}, \frac{\epsilon}{6} \right\}$ and our limit is 2. $\square$